

Design Pattern Identification and Evolution in a Django based Digital Document Management Application

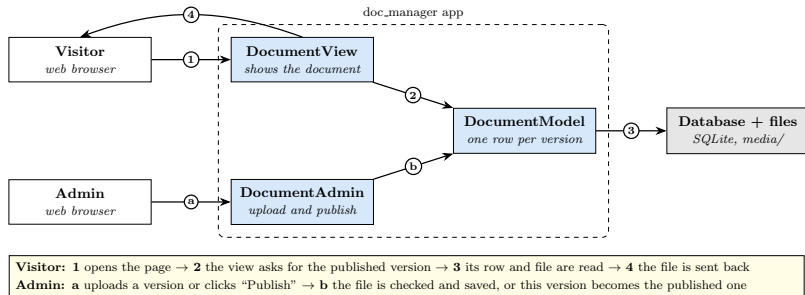
Team B5 — Raaghavan M S (AM.SC.U4CSE23330), V Sriman Vashishta
(AM.SC.U4CSE23368), CSE-D

23CSE455 Design Patterns — Case Study

2026-10-07

The app: `impicode/doc_manager` 1.0.2

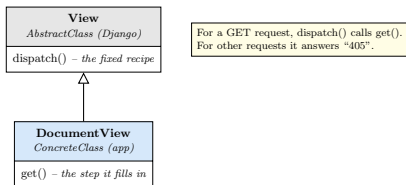
- ▶ A very small real Django app (MIT, 138 lines of code, 11 tests).
- ▶ An admin uploads versions of a document, e.g. a Privacy Policy (PDF or HTML).
- ▶ Old versions are kept; exactly one version is *published* for visitors.



The two patterns, in plain words

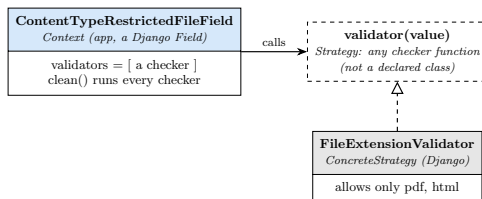
Template Method

A parent class owns a fixed *recipe*.
A child class fills in one step.



Strategy

An object keeps a *replaceable helper*
and calls it the same way every time.



Part 1 — what we found: 2 types, 4 instances

Instance	Pattern	Django part / app part
Published Document Page	Template Method	<code>View.dispatch</code> → <code>DocumentView.get</code>
Admin Read-only Fields	Template Method	<code>ModelAdmin</code> recipe → <code>get_readonly_fields</code>
Upload File Checks	Strategy	<code>validators</code> ← <code>FileExtensionValidator</code>
Upload Widget Choice	Strategy	<code>widget</code> ← <code>FileInput</code>

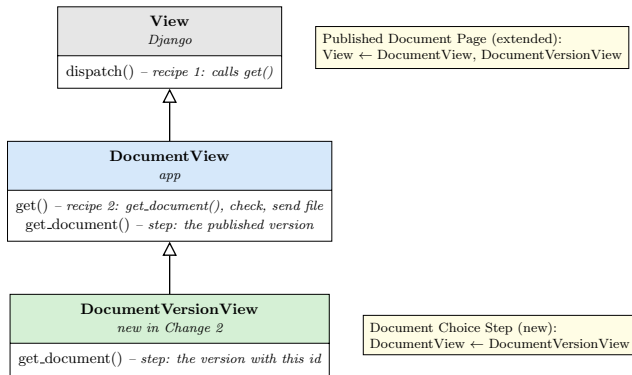
- ▶ Every instance is proven by a probe test that runs the real code (10/10).
- ▶ 7 candidates rejected (e.g. Singleton, Decorator, Factory Method), 0 uncertain.

Part 2 — three changes, each with a small refactoring first

Change	Refactoring	Pattern effect	Tests
CH01 accept .txt files	Extract Constant: one <code>FILE_TYPES</code> table	2 Strategy + 1 TM <i>modified</i>	11 → 19
CH02 open an older version	Extract Method <code>get_document()</code>	new TM “Document Choice Step”	19 → 22
CH03 lock a published file	Rename Parameter	admin hook <i>modified</i>	22 → 25

- ▶ New tests failed before each change and passed after; all logs are saved.
- ▶ Instances: 4 → 4 → 5 → 5. LOC 138 → 158. Highest CC 5 → 4.

Part 2 — a new Template Method appeared (CH02)



The new class changes only *one step*: which version to show (drafts stay hidden).

Part 3 — same problems in four languages

Language	Template Method			Strategy		
	LOC	CC	ms	LOC	CC	ms
Python	20	3	73.2	31	5	79.1
JavaScript	28	3	80.2	40	5	78.7
Java	30	3	201.8	51	5	224.7
C++	38	3	13.4	63	5	12.5

- ▶ All 8 programs pass the same shared cases (7/7 and 8/8).
- ▶ Java and C++ catch a missing step at compile time; Python and JavaScript at run time.

LLM use (Claude Opus 5.5) — checked claim by claim

Part	Correct	Partial	Wrong	What we used
1 discovery	11	1	2	a missed instance (proven first)
2 evolution	11	3	1	warning: CH02 showed drafts → fixed
3 languages	8	1	0	generated code tested only (8/8 pass)

- ▶ Prompts, inputs, raw outputs and our checks are saved for every run.

Findings

- ▶ Even a tiny Django app uses Template Method and Strategy: Django gives the fixed part, the app fills in the rest.
- ▶ The patterns made each change small: one table entry, one step, one hook.
- ▶ Refactoring first, with tests before and after, kept every change safe.
- ▶ Bugs found: the page crashed for other file types and the accept hint was ignored (both fixed); the admin search crashes (documented).